

Robotic Arms

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What Is a Robotic Arm?

- Robotic arms are devices that have been designed to do a given activity swiftly, correctly, and effectively
- They're usually motor-driven consisting of a collection of joints, articulations, and manipulators
- They are employed to accomplish heavy and/or repetitive processes quickly and consistently
- They are particularly valuable in the industries of industrial production, manufacturing, machining, and assembly



Advantages of Robotic Arms

- Improved Safety

Robotic arms keep employees safe by functioning in hazardous areas and doing jobs that pose a high risk of human injury.

- Better Productivity

Robotic arms can work nonstop for 24 hours a day, seven days a week, helping organisations enhance output by keeping production, inspections, and other operations running continually.

- Better Precision

Robotic arms, by their very nature, are more constant and precise than humans when it comes to jobs that need exceptional accuracy or consistency.

- Improved Flexibility

Robotic arms may readily be repurposed for new operations or installed on alternative platforms, such as autonomous mobile robots (AMRs), a fixed assembly line platform, or a wall or shelf, when business objectives change.

Why Are Industrial Robotic Arms Used in Manufacturing?

Industrial robotic arms are used in production for several purposes:

- Where it is necessary to reduce threats to humans
- When greater strength or precision is required than a person
- When the task is too monotonous for human employees
- When the continuous operation is needed

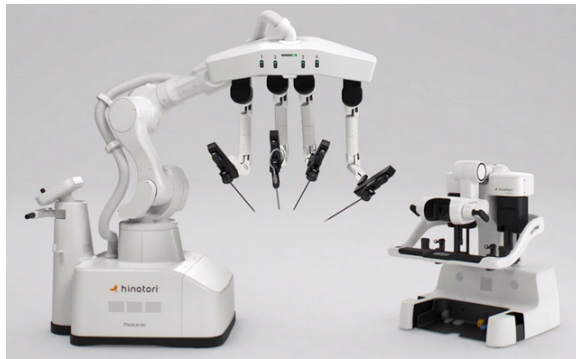
Industrial robots



Industrial robots are mainly used in:

- manufacturing
- logistics
- agriculture
- animal husbandry

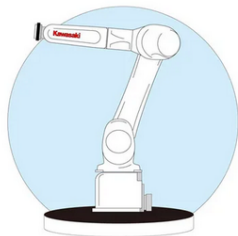
Non-industrial robots



Non-industrial robots are mainly used in:

- security
- medical and health care
- space

Conventional vs collaborative robots



Characteristics of Conventional industrial robots

- 1 Can work with heavy objects, high speed, and high precision
- 2 Can handle dull, dirty, and dangerous work
- 3 Resilient and able to work 24 hours a day



Characteristics of Collaborative robots

- 1 Space saving
- 2 Can be introduced at low cost
- 3 Flexibility to work in various manufacturing sites

- Conventional industrial robots are required to be installed away from people and inside safety fences
- Collaborative robots can coexist with humans and can be operated in a small space without the need for large safety fences

World record of 500,000 units exceeded

Annual installations of industrial robots - World

1,000 units

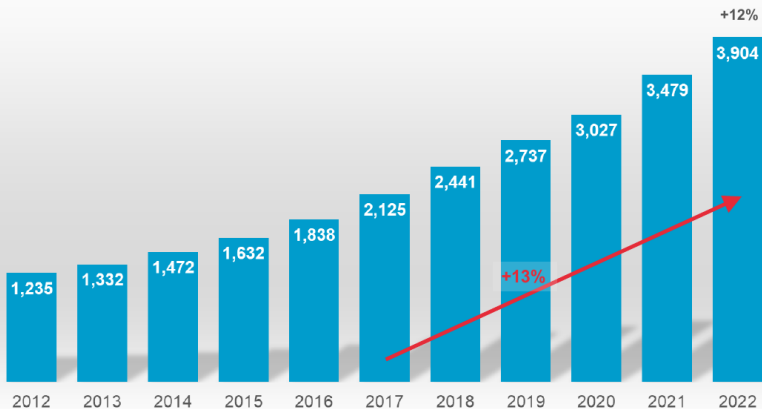


Source: World Robotics 2023

Almost **4 million** industrial robots **operating** around the world

Operational stock of industrial robots - World

1,000 units

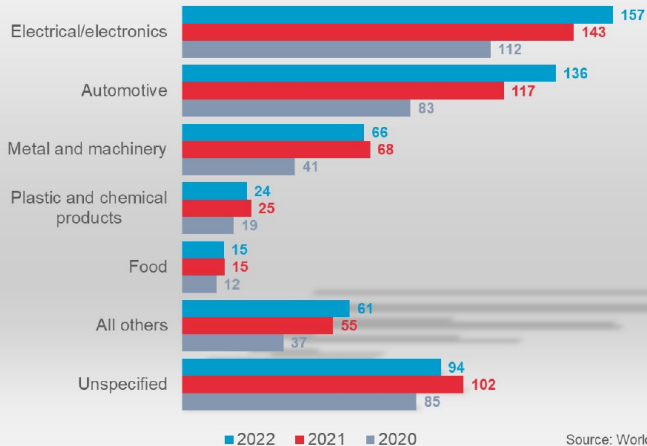


Source: World Robotics 2023

Electronics is major customer – challenges for general industry

Annual installations of industrial robots by customer industry - World

1,000 units

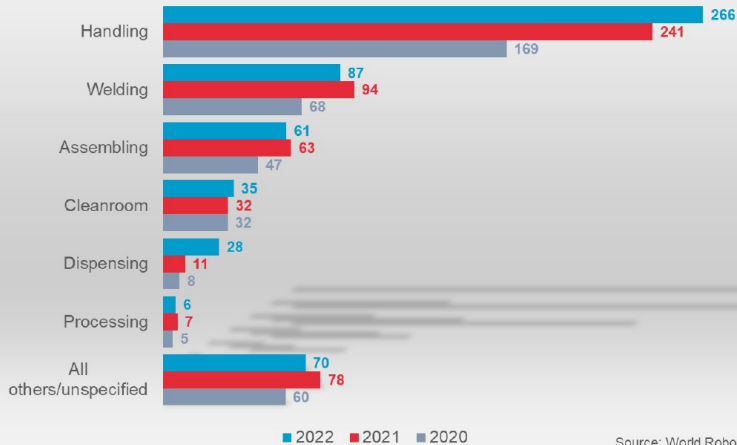


Source: World Robotics 2023

Handling is most important application with 48% share

Annual installations of industrial robots by application - World

1,000 units

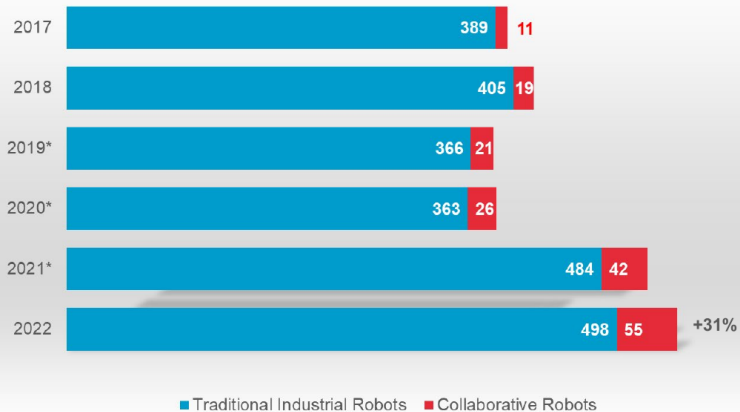


Source: World Robotics 2023

Collaborative robots steadily growing their market share to 10%

Collaborative and traditional industrial robots

'000 units

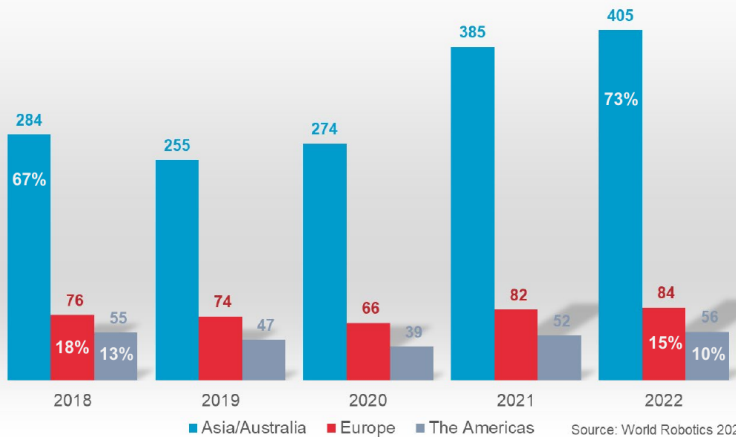


*revised

Source: World Robotics 2023

Growth in all regions

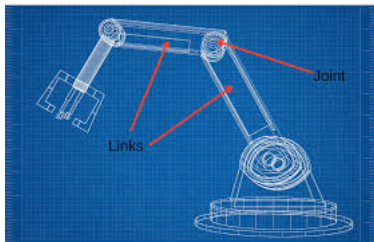
Annual installations of industrial robots
('000 of units)



Robot Mechanical Structure

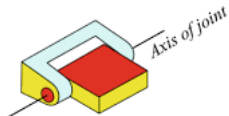
Robot Manipulators

- The mechanical structure of a robot manipulator consists of a sequence of rigid bodies (**links**) interconnected by means of articulations (**joints**).
- A manipulator is characterized by:
 - an **arm** that ensures mobility;
 - a **wrist** that confers dexterity;
 - an **end-effector** that performs the task required of the robot.
- The fundamental structure of a manipulator is the serial or open **kinematic chain**: the sequence of links connecting the two ends of the chain.

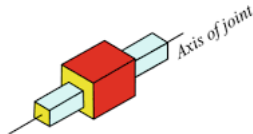


Joints

- A manipulator's mobility is ensured by the presence of joints.
- The articulation between two consecutive links can be realized by means of either a **prismatic** or a **revolute** joint.
- In an open kinematic chain, each prismatic or revolute joint provides the structure with a single degree of freedom (DOF).
- A prismatic joint creates a relative translational motion between the two links.
- A revolute joint creates a relative rotational motion between the two links.
- Revolute joints are usually preferred to prismatic joints in view of their compactness and reliability.



Revolute joint



Prismatic joint

Video revolute Video prismatic

Degrees of freedom

- The degrees of freedom should be properly distributed along the mechanical structure in order to have a sufficient number to execute a given task.
- In the most general case of a task consisting of arbitrarily positioning and orienting an object in three-dimensional (3D) space, six DOFs are required:
 - three for positioning a point on the object
 - three for orienting the object with respect to a reference coordinate frame
- If more DOFs than task variables are available, the manipulator is said to be redundant from a kinematic viewpoint.

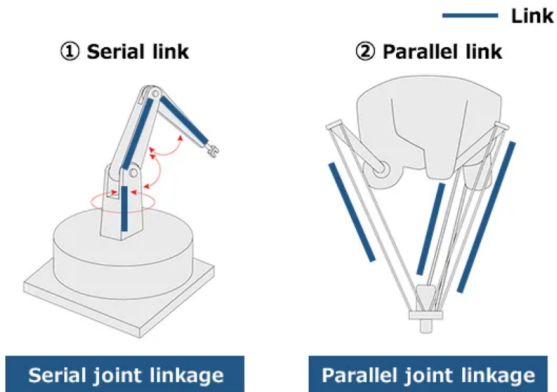
- Workspace.
 - The workspace represents that portion of the environment the manipulator's end-effector can access.
 - Its shape and volume depend on the manipulator structure as well as on the presence of mechanical joint limits.
- Load Capacity
 - Load capacity, a primary robot specification, is closely coupled with acceleration and speed

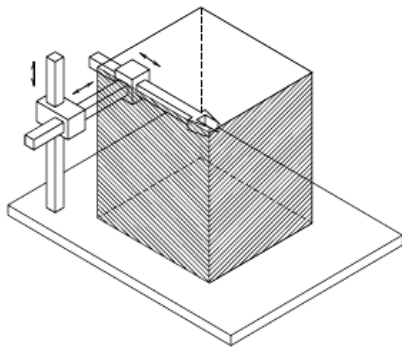
The Different Types of Robotic Arms

Serial vs Parallel link

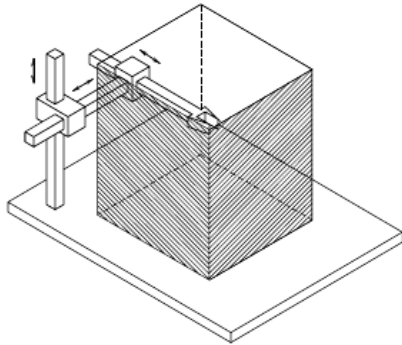
Robots are roughly categorized into two types according to how their links are arranged:

- Serial link
- Parallel link

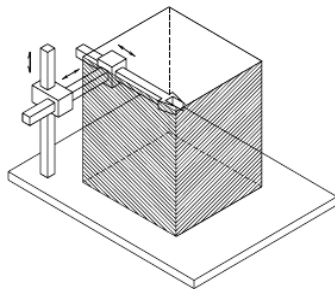




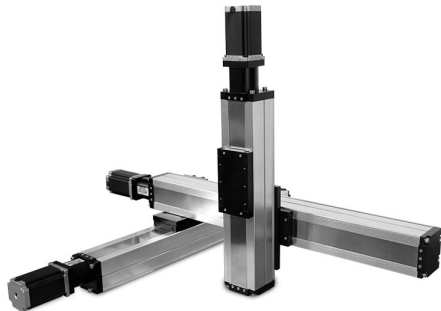
- Cartesian geometry is realized by three prismatic joints whose axes typically are mutually orthogonal.
- In view of the simple geometry, each DOF corresponds to a Cartesian space variable and thus it is natural to perform straight motions in space.



- The Cartesian structure offers very good mechanical stiffness.
- Wrist positioning accuracy is constant everywhere in the workspace.
- This is the volume enclosed by a rectangular parallel-piped.
- As opposed to high accuracy, the structure has low dexterity since all the joints are prismatic.

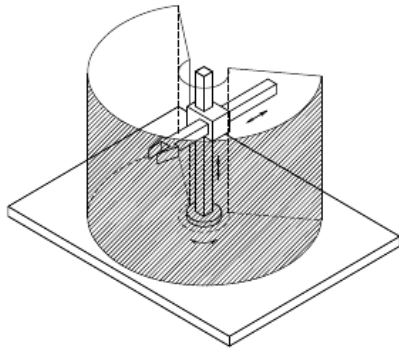


- Such a structure makes available a workspace with a large volume and enables the manipulation of objects of large dimensions and heavy weight.
- Cartesian manipulators are employed for material handling and assembly.
- The motors actuating the joints of a Cartesian manipulator are typically electric and occasionally pneumatic.



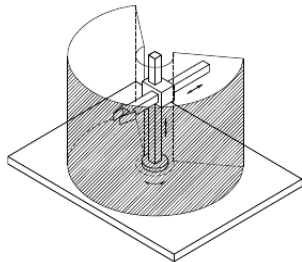
Video

Cylindrical - 1



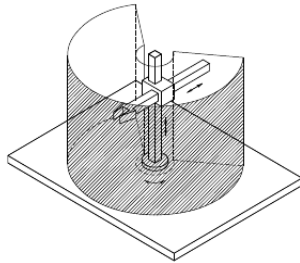
- Cylindrical geometry differs from Cartesian in that the first prismatic joint is replaced with a revolute joint.
- If the task is described in cylindrical coordinates, in this case each DOF also corresponds to a Cartesian space variable.

Cylindrical - 2



- The cylindrical structure offers good mechanical stiffness.
- Wrist positioning accuracy decreases as the horizontal stroke increases.
- The workspace is a portion of a hollow cylinder.
- The horizontal prismatic joint makes the wrist of a cylindrical manipulator suitable to access horizontal cavities.

Cylindrical - 3



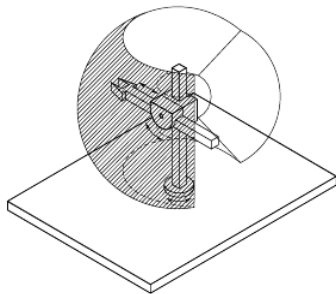
- Cylindrical manipulators are mainly employed for carrying objects even of large dimensions
- In such a case the use of hydraulic motors is to be preferred to that of electric motors.

Cylindrical - 4

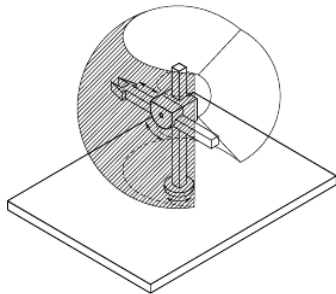


Video

Spherical - 1

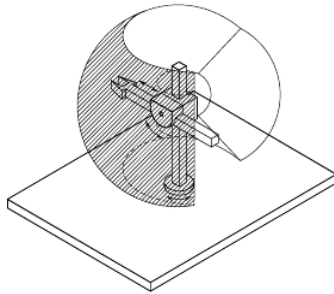


- Spherical geometry differs from cylindrical in that the second prismatic joint is replaced with a revolute joint.
- Each DOF corresponds to a Cartesian space variable provided that the task is described in spherical coordinates.



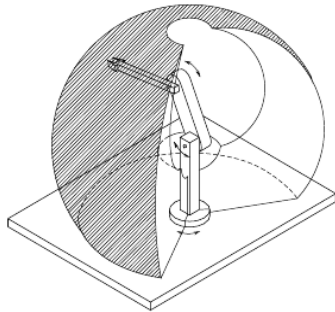
- Mechanical stiffness is lower than the above two geometries and mechanical construction is more complex.
- Wrist positioning accuracy decreases as the radial stroke increases.
- The workspace is a portion of a hollow sphere

Spherical - 3

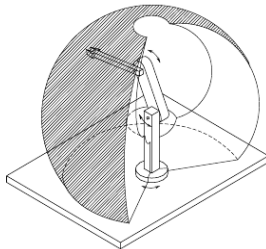


- Spherical manipulators are mainly employed for machining.
- Electric motors are typically used to actuate the joints.

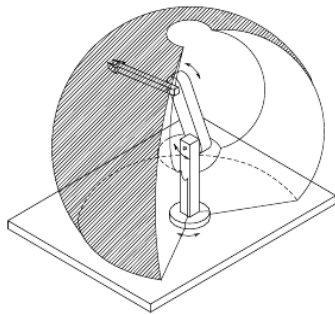
Anthropomorphic - 1



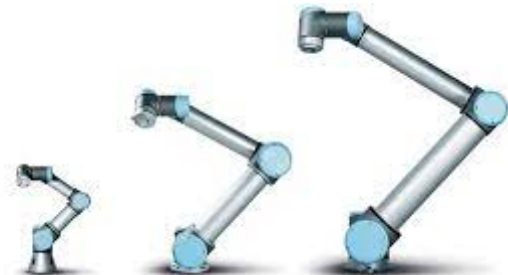
- Anthropomorphic geometry is realized by three revolute joints
- The revolute axis of the first joint is orthogonal to the axes of the other two which are parallel.
- By virtue of its similarity with the human arm, the second joint is called the shoulder joint and the third joint the elbow joint since it connects the “arm” with the “forearm.”



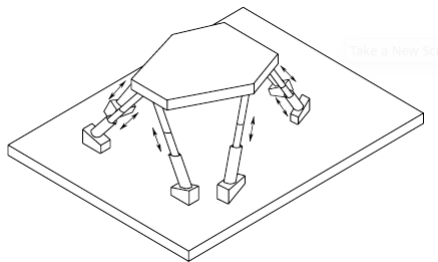
- The anthropomorphic structure is the most dexterous one, since all the joints are revolute.
- On the other hand, the correspondence between the DOFs and the Cartesian space variables is lost
- Wrist positioning accuracy varies inside the workspace.
- Workspace is approximately a portion of a sphere and its volume is large compared to manipulator encumbrance. Joints are typically actuated by electric motors. The range of industrial applications of anthropomorphic manipulators is wide.



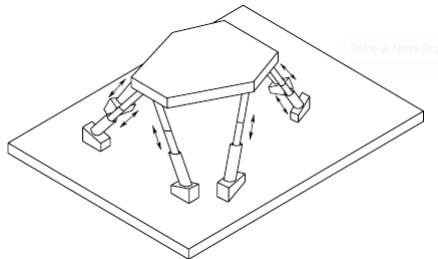
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Video

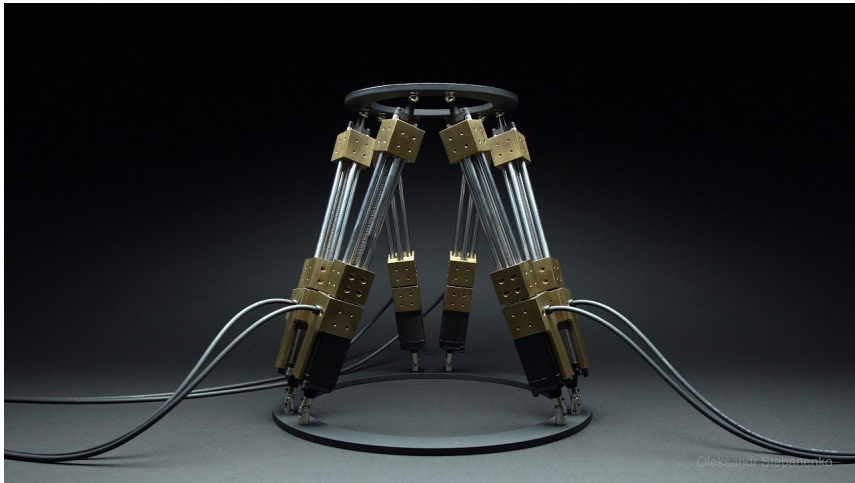


- All the previous manipulators have an open kinematic chain.
- Whenever larger payloads are required, the mechanical structure will have higher stiffness to guarantee comparable positioning accuracy.
- In such a case, resorting to a closed kinematic chain is advised.
- An interesting closed-chain geometry is parallel geometry which has multiple kinematic chains connecting the base to the end-effector.

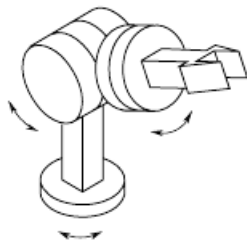


- The fundamental advantage is seen in the high structural stiffness, with respect to open-chain manipulators, and thus the possibility to achieve high operational speeds.
- The drawback is that of having a reduced workspace.

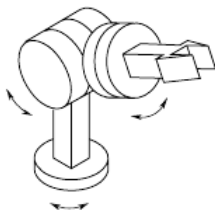
Parallel - 3



Video



- The manipulator structures previously presented are required to position the wrist which is then required to orient the manipulator's end-effector.
- If arbitrary orientation in 3D space is desired, the wrist must possess at least three DOFs provided by revolute joints.



- Spherical wrist: the three revolute axes intersect at a single point.
- The key feature of a spherical wrist is the decoupling between position and orientation of the end-effector:
 - the arm is entrusted with the task of positioning the above point of intersection
 - the wrist determines the end-effector orientation
- Those realizations where the wrist is not spherical are simpler from a mechanical viewpoint, but position and orientation are coupled, and this complicates the coordination between the motion of the arm and that of the wrist to perform a given task.



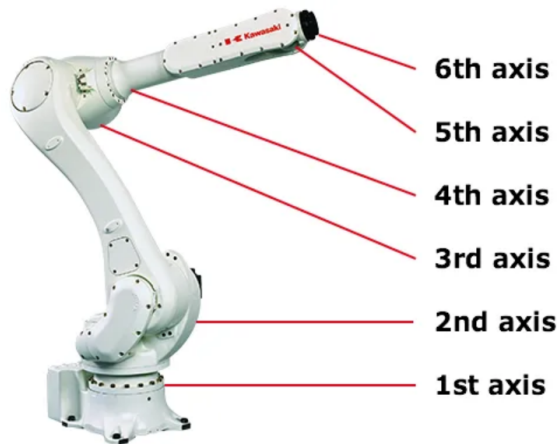
- The end-effector is specified according to the task the robot should execute.
- For material handling tasks, the end-effector consists of a gripper of proper shape and dimensions determined by the object to be grasped.
- For machining and assembly tasks, the end-effector is a tool or a specialized device, e.g., a welding torch, a spray gun, a mill, a drill, or a screwdriver.

End-effector - 2



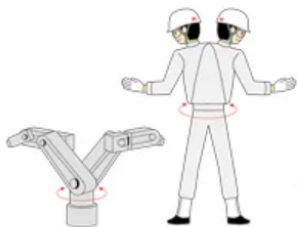
Robot and human motion comparison

- A vertical articulated robot is an industrial robot with a serial link structure
- It is generally composed of six joints (6 axes)

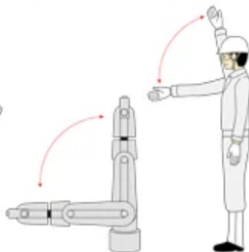


Robot and human motion comparison - 2

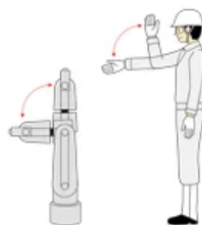
1st axis: Waist



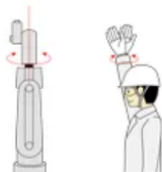
2nd axis: Shoulder



3rd axis: Elbow



4th axis: Wrist (Rotating)



5th axis: Wrist (Bending)



6th axis: Fingertip



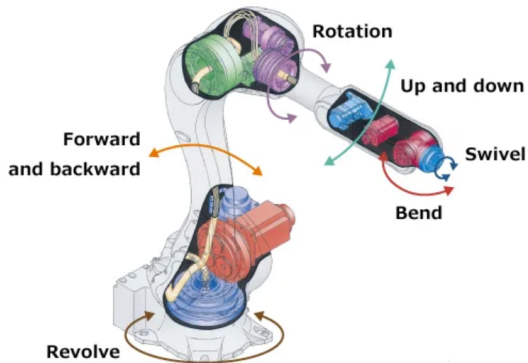
Video

What is required to move joints?

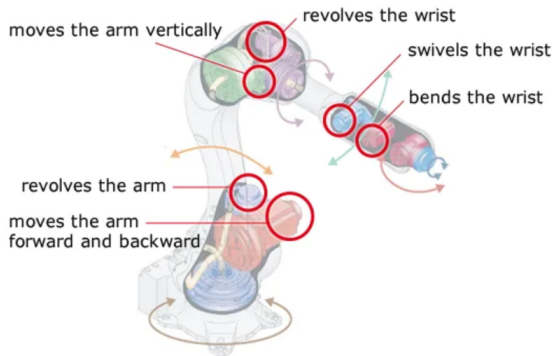
Internal structure of industrial robots

A robot is made up of many different parts

- actuator
- reduction gear
- encoder
- transmission



Actuator



- The actuator is a component that functions as the joint of the robot
- It allows a robot to move the arm up and down or rotate
- It converts energy into mechanical motions

Reduction gear

- A reduction gear is a device for increasing the power of a motor
- A motor alone is limited in the amount of power it can output
- In order to generate great power, motors are basically used in combination with this reduction gear



Spur Gear Set



Straight gear Set



Spiral Bevel Gear



Worm Gear Set



Helical Gear Set



Planetary Gear Set



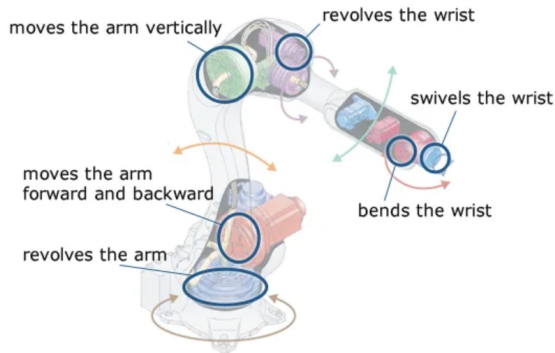
Herringbone Gear



Hypoid Gear

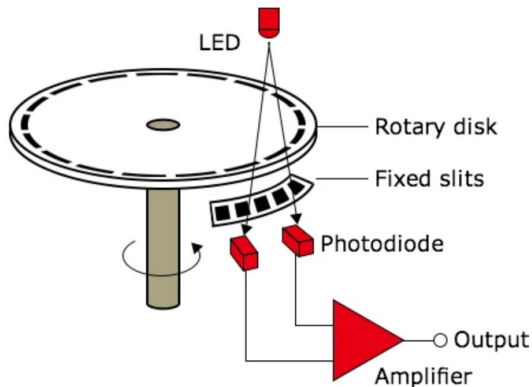
Reduction gear - 2

- If you combine gear wheels with the different number of gears and reduce the motor's rotation by a factor of 10, the power of the motor will be multiplied by 10



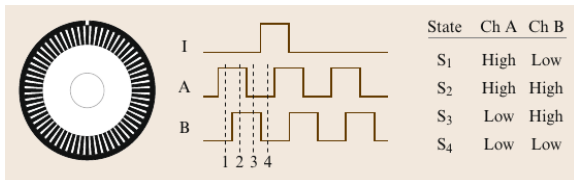
Encoder

- An encoder is a device that indicates the position (angle) of a motor's rotational shaft
- Having an encoder, it can provide tangible data about in what direction and how much the robot moves



Encoder - 2

- General optical encoders have a disk attached to the rotating shaft of the motor
- The disk has slits at regular intervals to let light pass and on both sides of the disk are light-emitting-diode (LED) and light receiving elements (photodiodes) to discriminate the light intensity (light and dark)
- When the motor rotates, the light either passes through the slits or is blocked, so the rotation angle and speed can be determined by reading the signals
- This allows servo motors to precisely control positioning and speed.



Video 1

Video 2

Transmission

- The transmission is a component that transmits the power generated by the actuators and reduction gears
- The transmission is also capable of changing the direction and magnitude of power



Video

Controllers

- The controllers are the robotic arms' main processors and operate as their brains
- They can be set to behave automatically, or they can be manually operated by receiving instructions from a specialist
- They are the control consoles for mechanical arms, and they come in several forms depending on the amount of computing power required.



- Mask Manufacturing
- Screw tightening
- Robotic House Construction
- Welding and cutting of large steel materials
- Grinding and polishing